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Driss Bouggar

Universite Ibn Tofail Kenitra Faculte des Sciences

Mohamed EL Fatini

Universite Ibn Tofail Kenitra Faculte des Sciences

Bouchra R. Nasri

Université de Montréal: Universite de Montreal

Idriss Sekkak

idrisssekkak@gmail.com

University of Montreal: Universite de Montreal <https://orcid.org/0000-0002-5878-8369>

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Probability analysis and near-optimal control for a stochastic epidemic treatment model with relapse

Driss Bouggar · Mohamed El Fatini · Bouchra
R. Nasri · Idriss Sekkak.

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Abstract Epidemiology is a field for evaluating the sources of illness and clinical effects in communities. Therefore, individuals are viewed globally in order to investigate the dynamics and the control of an infectious disease. However, any governmental spending on a pharmaceutical control policy in times of epidemics can lead to a state budget imbalance. For this matter, we study in this paper stochastic stability and optimization for a treatment strategy, where the infectious disease is characterized by the relapse aspect. We provided a stochastic threshold for extinction and persistence in mean. Then, we established a priori estimates of the optimal control, where we obtained sufficient and necessary conditions for the considered treatment problem of the stochastic epidemic model by applying the maximum condition on the Hamiltonian function and the Ekeland principle. We provide a characterization of stochastic optimal control and sensitivity analysis to discuss its impact on the basic reproduction number with numerical illustrations for the system to carry out our theoretical investigations.

Keywords Near optimal control · Epidemic model · Relapse · Stochastic process · White noise · Sensitivity analysis · Stability analysis.

1 Introduction

In the last decade, epidemics such as COVID-19 and Ebola diseases have resulted in a devastating loss of lives across the globe and they pose an immense threat to public health, food security, governmental spending, and the corporate world.

Driss Bouggar & Mohamed El Fatini
Laboratory of PDE's, Algebra and Spectral Geometry.
Faculty of Sciences, Ibn Tofail University, BP 133, Kénitra, Morocco.
E-mail: drissbouggar@gmail.com & melfatini@gmail.com

Bouchra R. Nasri & Idriss Sekkak
Centre de Recherche en Santé Publique and Département de médecine sociale et préventive,
École de santé publique, Université de Montréal,
7101 Park Ave, Montreal, Quebec H3N 1X9, Canada.
E-mail: bouchra.nasri@umontreal.ca & idrisssekkak@gmail.com

Therefore, those events shed the light on mathematical modeling of infectious diseases in order to build efficient policies to prevent and control the disease's spread. Although, reverting back propriety of infectious disease, with environmental exposure, was not extensively investigated, which characterizes diseases like herpes [1] and tuberculosis [2]. For this matter, Tudor [3] has formulated a SIRS epidemic model, where individuals subject to a reactivation of the latent infection or inadequate care, recovered population can revert back to the infective class. In the literature, several authors have explored the deterministic dynamics [4, 5]. In addition, the incorporation of environmental effects for a transmission rate has been studied by El Fatini et al [6]. Thereafter, the impact of media coverage was discussed by Caraballo et al [7] for a stochastic SIRS epidemic model, while the stochastic epidemic model with jumps was investigated by Berrhazi et al. [8, 9]. That being said, a quantitative and qualitative analysis is insufficient, and managing an infectious disease using strategies, such as vaccines and care, can be very costly, resulting in increased government spending.

Therefore, using control theory, many researchers included the control variable to describe the vaccination [10, 11] and treatment [12, 13] strategies. For a stochastic model, the nonuniqueness of an existent optimal control is a very challenging problem to explore. Hence, as a feasible possibility, many authors started to explore a near-optimal control problem for epidemic models [14–16], viral [17] and chemostat [18] models, where the hypothesizes are weaker.

Therefore, we investigate a controlling approach for the following stochastic epidemic model [6] with relapse

$$\begin{aligned} dS(t) &= [\mu - \mu S(t) - \beta S(t)I(t)]dt - \sigma S(t)I(t)dB(t), \\ dI(t) &= [\beta S(t)I(t) - (\mu + \lambda)I(t) + \rho R(t)]dt + \sigma S(t)I(t)dB(t) \\ dR(t) &= [\lambda I(t) - (\mu + \rho)R(t)]dt, \end{aligned} \quad (1)$$

where the parameters of the system (1) describe the follows: β presents the disease transmission rate. μ denotes the recruitment rate of susceptible individuals corresponding to the new individuals coming to the population in system (1), such as new births and immigrants and also the leaving or dead individuals, ρ describes the relapse rate. λ denotes the recovery rate from an infectious disease, σ indicates the intensity of a continuous environmental noise, and $B(t)$ is an independent standard Brownian motion, where the complete probability space denoted by $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ with a filtration $\{\mathcal{F}_t\}_{t \geq 0}$ satisfying the usual conditions. Besides, epidemic models are characterized by a basic reproduction number, which denotes the average number of transmissions of a single infectious individual on the susceptible size.

$$R_0 = \frac{\beta}{\mu + \lambda - \frac{\rho\lambda}{\mu + \rho}}$$

The analysis of the deterministic model for (1) has been carried out by Georgescu and Zhang [19], where they proved the solutions behave according to the threshold R_0 such that, If $R_0 < 1$, the deterministic part of the system (1) has a unique free-disease equilibrium $E_0(1, 0, 0)$. Otherwise, if $R_0 > 1$; the deterministic part of the system (1) has a unique positive equilibrium [19] $E^*(S^*, I^*, R^*)$.

In [20], authors proposed an optimal control for both deterministic and stochastic models with a Crowley-martin functional response. Although, our motivation is to

seek fewer restrictions with a near optimality approach for the stochastic epidemic model (1), since an optimal control tends to be very sensitive to outer perturbations and it does not give an opportunity to adjust to new circumstances.

Therefore, the controlled stochastic epidemic model corresponding to the system (1) can be expressed as follows

$$\begin{aligned} dS(t) &= [\mu - \mu S(t) - \beta S(t)I(t)]dt - \sigma S(t)I(t)dB(t) \\ &= b_1(x(t))dt + \sigma_1(x(t))dB(t), \\ dI(t) &= [\beta S(t)I(t) - (\mu + \lambda)I(t) + \rho R(t) - mu(t)I(t)]dt + \sigma S(t)I(t)dB(t) \quad (2) \\ &= b_2(x(t), u(t))dt + \sigma_2(x(t))dB(t), \\ dR(t) &= [\lambda I(t) + mu(t)I(t) - (\mu + \rho)R(t)]dt \\ &= b_3(x(t), u(t))dt, \end{aligned}$$

where m is a positive constant. Let $U \subset \mathbb{R}$ represents a bounded nonempty closed set, and $u : [0, T] \times \Omega \rightarrow U$ is an $\mathcal{F}_t$ -adapted process, which it presents fitted and timely pharmaceutical interventions acting on decreasing the impact of disease on the infected population. Later, we will use $\mathcal{U}_{ad}$ to describe all admissible controls.

The cost function, related to the stochastic epidemic model (2) with relapse, is represented as follows

$$J(0, x_0, u(t)) = \mathbb{E} \left\{ \int_0^T l(t, x(t), u(t))dt + g(x(T)) \right\}, \quad (3)$$

where $x(t) = (x_1(t), x_2(t), x_3(t))^\top = (S(t), I(t), R(t))^\top$

The cost of treatment policy is represented by the function $l(t, x(t), u(t))$ and the function $g(x(T))$ for the size of the population, with, $l : [0, T] \times \mathbb{R}^3 \times \mathcal{U}_{ad} \rightarrow \mathbb{R}$, $g : \mathbb{R}^3 \rightarrow \mathbb{R}$ are measurable.

The main preoccupation of the control problem is to look for an admissible control that has a near minimum for the objective function $J(0, X_0, u(t))$ over $u(t) \in \mathcal{U}_{ad}$. The value function is defined as

$$v(0, x_0) = \min_{u(t) \in \mathcal{U}_{ad}} J(0, x_0, u(t)).$$

The paper is organized as follows. In Section (2), we handle some basic notions on the near-optimal control and the stochastic epidemic model (1). In Section (3), we provide a stochastic threshold for the controlled system (2) to determine the extinction and persistence in mean of an infectious disease with relapse. In Section (4), we explore some assumptions in order to investigate sufficient and necessary conditions for the near-optimal control of the stochastic epidemic model (2). In Section (5), we study a characterization of stochastic optimal control and we illustrate numerically some scenarios for a treatment. In section (6), we provide a sensitivity analysis for the stochastic model with respect to optimal control. Finally, we provide a conclusion on the investigation.

2 Background materials

Throughout this paper, We investigate a stochastic near-optimal treatment problem for a stochastic epidemic model with relapse (2) on a finite time horizon $[0, T]$.

the complete filtered probability space is denoted by $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$. we assume that the following hypotheses are fulfilled.

- (H_1) For any $t \in [0, T]$, the partial derivatives $f_{x(t)}(x(t), u(t))$ and $g_{x(t)}(x(t))$ are continuous and there exists a constant C such that

$$|f_{x(t)}(x(t), u(t))| \leq C(1 + |x(t)|), \quad (1 + |x(t)|)^{-1} |g_{x(t)}(x(t))| \leq C.$$

- (H_2) $\forall x(t), x'(t) \in \mathbb{R}_+^3$ and $u(t), u'(t) \in \mathcal{U}_{ad}$, for any $0 \leq t \leq T$, the function $f(x(t), u(t))$ is differentiable in $u(t)$. There also exists a constant C such that

$$|g_{x(t)}(x(t)) - g_{x(t)}(x'(t))| \leq C|x(t) - x'(t)|,$$

$$|f(x(t), u(t)) - f(x(t), u'(t))| + |f_{u(t)}(x(t), u(t)) - f_{u'(t)}(x(t), u'(t))| \leq C|u(t) - u'(t)|.$$

Since in epidemic modeling, the population can be described only with positive solutions. Therefore, authors in [6] used the Lyapunov analysis method to prove that the stochastic epidemic model (2) has a local positive solution in Γ , then they showed the global solution. Moreover, the set Γ is considered as

$$\Gamma = \{(S, I, R) \in \mathbb{R}_+^3 : S + I + R = 1\}$$

As a result, the following theorem verifies the existence and uniqueness of the global positive solution

Theorem 1 [6] *For any given initial value $(S(0), I(0), R(0)) \in \mathbb{R}_+^3$, there exists a unique global positive solution of the stochastic model (2) on $t \geq 0$ and the solution remains in Γ with probability 1, almost surely.*

In the following, we will give the definitions of an optimal control, a ε -optimal control and a near-optimality as stated in [22]

Definition 1 [23](Optimal control)

An admissible control $u^*(t) \in \mathcal{U}_{ad}[0, T]$, is called optimal control, if

$$J(u^*(t)) = \min_{u(t) \in \mathcal{U}_{ad}[0, T]} J(u(t))$$

Definition 2 (ε -optimal control)

For a given $\varepsilon > 0$, an admissible pair $(x^\varepsilon(t), u^\varepsilon(t))$, or an admissible control $u^\varepsilon(t)$ is called ε -optimal if

$$|J(0, x_0, u^\varepsilon(t)) - v(0, x_0)| \leq \varepsilon.$$

Definition 3 (Near-optimal)

A family of admissible controls $\{u^\varepsilon(t)\}$ parameterized by $\varepsilon > 0$ and any element $u^\varepsilon(t)$, in the family, are called near-optimal, if

$$|J(0, x_0, u^\varepsilon(t)) - v(0, x_0)| \leq \delta(\varepsilon).$$

For sufficient small ε , where δ is a function of ε satisfying $\delta(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$. The estimate $\delta(\varepsilon)$ is called an error bound. If $\delta(\varepsilon) = c\varepsilon^l$ for some $c, l > 0$, then $u^\varepsilon(t)$ is called near-optimal with order ε^l

Definition 4 ([24]) Let $E \subset \mathbb{R}^n$ be a region and $\phi : E \rightarrow \mathbb{R}^n$ be a locally Lipschitz continuous function. The generalized gradient of ψ at $v \in E$ is defined as

$$\partial\psi(v) = \left\{ v \in \mathbb{R}^n \mid \langle v, \theta \rangle \leq \lim_{y \rightarrow v, y \in E, h \downarrow 0} \frac{\psi(y + h\theta) - \psi(y)}{h} \right\}. \quad (4)$$

Next, we present Ekeland's principle, which verifies the near optimality of solutions for optimization problems.

Lemma 1 *Ekeland's principle* [25] Let $(Q, \tilde{d})$ be a complete metric space and $F(\cdot) : Q \rightarrow \mathbb{R}$ be a lower-semicontinuous and bounded from below. For any $\varepsilon > 0$, we assume that $u^\varepsilon \in Q$ satisfies

$$F(u^\varepsilon) \leq \inf_{u \in Q} F(u) + \varepsilon.$$

Then there is a $u^\theta \in Q$ such that for all $\theta > 0$ and $u \in Q$

$$F(u^\theta) \leq F(u^\varepsilon), \quad \tilde{d}(u^\theta, u^\varepsilon) \leq \theta, \quad \text{and} \quad F(u^\theta) \leq F(u) + \frac{\varepsilon}{\theta} \tilde{d}(u, u^\theta). \quad (5)$$

In this work, we consider also a normalized forward sensitivity index of a variable corresponding to a parameter, where it represents the proportion of a relative change in the variable regarding a relative change in the parameter when the variable is denoted as a differentiable function for a parameter, the index can be described using partial derivatives. Hence, the normalized forward sensitivity index is defined as follows

Definition 5 [28] The normalized forward sensitivity index of a variable u that depends differentiably on a parameter X is defined as:

$$\phi_X^u = \frac{\partial u}{\partial X} \frac{X}{|u|}. \quad (6)$$

3 Stability analysis of a controlled stochastic model

The condition for infectious disease eradication and persistence in the stochastic epidemic model (2) is highlighted in this section. Therefore, we consider the stochastic threshold as

$$\mathcal{R}_s(t) = \mathcal{R}_0(t) - \frac{\sigma^2}{2 \left(\mu + \lambda + mu(t) - \frac{\rho\lambda}{\mu+\rho} \right)},$$

where

$$\mathcal{R}_0(t) = \frac{\beta}{\mu + \lambda + mu(t) - \frac{\rho\lambda}{\mu+\rho}}.$$

In the following, we denote throughout this investigation $\mathcal{R}_s(t)$ as $\mathcal{R}_s$.

3.1 Extinction of the infectious disease:

Theorem 2 *If $\mathcal{R}_s < 1$, then for any initial condition $(S_0, I_0, R_0) \in \Gamma$, we have*

$$\lim_{t \rightarrow \infty} (S(t), I(t), R(t)) = E_0 \quad a.s. \quad (7)$$

Proof We prove that for any $\varepsilon > 0$, we have $\varepsilon_1 > 0$ such that

$$\mathbb{P} \left\{ \lim_{t \rightarrow \infty} (S(t), I(t), R(t)) = E_0 \right\} \geq 1 - \varepsilon \quad (8)$$

For all $(S_0, I_0, R_0) \in \mathcal{O}_{\varepsilon_1} := (1 - \varepsilon_1, 1) \times (0, \varepsilon_1)^2$. Let consider

$$dS(t) = F_1(X)dt - G(X)dW(t), \quad dI(t) = F_2(X)dt + G(X)dW(t). \quad (9)$$

By calculating the derivatives of F_1, F_2 , and G at the equilibrium point E_0 , we obtain

$$\frac{\partial F_1}{\partial S}(E_0) = -\mu, \quad \frac{\partial F_1}{\partial I}(E_0) = -\beta, \quad \frac{\partial F_1}{\partial R}(E_0) = 0,$$

$$\frac{\partial F_2}{\partial S}(E_0) = 0, \quad \frac{\partial F_2}{\partial I}(E_0) = \beta - (\mu + \lambda + mu(t)), \quad \frac{\partial F_2}{\partial R}(E_0) = \rho,$$

$$\frac{\partial G}{\partial S}(E_0) = \frac{\partial G}{\partial R}(E_0) = 0, \quad \frac{\partial G}{\partial I}(E_0) = \sigma. \quad (10)$$

Applying Taylor expansions of F_1, F_2 , and g in E_0 , we get

$$\begin{aligned} F_1(X) &= \mu - \mu S - \beta I + o, \\ F_2(X) &= [\beta - (\mu + \lambda + mu(t))] I + \rho R + o, \\ G(X) &= \sigma I + o, \end{aligned} \quad (11)$$

with

$$o := o \left(\sqrt{(S-1)^2 + I^2 + R^2} \right),$$

and

$$\lim_{X \rightarrow E_0} \frac{o \left(\sqrt{(S-1)^2 + I^2 + R^2} \right)}{\sqrt{(S-1)^2 + I^2 + R^2}},$$

Now, we define $V_2(X) = \left(I + \frac{\rho}{\mu + \rho} R \right)^\nu$, such that ν is a positive constant.

Applying Itô's formula, we obtain

$$\begin{aligned} \mathcal{L}V_1(X) &= \nu \left(I + \frac{\rho}{\mu + \rho} R \right)^{\nu-1} \left(\beta - \mu - \lambda - mu(t) + \frac{\rho\lambda}{\mu + \rho} \right) I \\ &\quad + \left(I + \frac{\rho}{\mu + \rho} R \right)^{\nu-2} \nu(\nu-1) \frac{\sigma^2}{2} I^2 \\ &\leq \nu \left(I + \frac{\rho}{\mu + \rho} R \right)^{\nu-2} I \left[\beta - \mu - \lambda - mu(t) + \frac{\rho\lambda}{\mu + \rho} + \frac{(\nu-1)\sigma^2}{2} \right] \end{aligned} \quad (12)$$

If $\mathcal{R}_s < 1$, we get that $\mathcal{L}V_1(X) \leq 0$. Hence, we obtain

$$\lim_{t \rightarrow \infty} I(t) = 0. \quad (13)$$

Let $\Omega = \{\omega \in \Omega : \lim_{t \rightarrow \infty} I(t) = 0\}$, then it implies $P(\Omega) = 1$. Therefore, for any $\omega \in \Omega$ and any constant $\varepsilon > 0$, there exists a constant $T(\omega, \varepsilon) > 0$ such that $I(\omega, t) \leq \varepsilon$ for all $t > T$. From (2), we get

$$dR(\omega, t) \leq \lambda\varepsilon - (\mu + \rho)R(\omega, t)dt \quad \text{for all } \omega \in \Omega, t > T,$$

which implies by comparison theorem that $\limsup_{t \rightarrow \infty} R(\omega, t) \leq \frac{\lambda\varepsilon}{\mu + \rho}$ for all $\omega \in \Omega$. Noting that $R(\omega, t) > 0$ for all $\omega \in \Omega$ and $t > 0$, by arbitrariness of ε , we have

$$\lim_{t \rightarrow \infty} R(\omega, t) = 0, \omega \in \Omega.$$

Recalling that $P(\Omega) = 1$, we obtain that

$$\lim_{t \rightarrow \infty} R(t) = 0, a.s. \quad (14)$$

Finally, from (13), and (14), we get that

$$\lim_{t \rightarrow \infty} S(t) = 1, a.s. \quad (15)$$

Hence, we obtain that the disease free-equilibrium E_0 is locally asymptotically stable, which proves (8). Next, we prove that if $\mathcal{R}_s < 1$ for any initial condition $(S_0, I_0, R_0) \in \Gamma$, the solution $(S(t), I(t), R(t))$ will eventually enter $\mathcal{O}_{\varepsilon_1}$. Let define the following stopping time

$$\tau_{\varepsilon_1} = \inf\{t \geq 0 : S(t) \geq 1 - \varepsilon_1\}, \quad (16)$$

and determine the following non-negative function C^2 -function $V_3(X) = a_1 - (S + 1)^{a_2}$, where $a_1 =$, and $a_2 > 2$. Applying Itô's formula, we get

$$\mathcal{L}V_3 \leq -a_2(S + 1)^{a_2-2} \left((S + 1)(\mu - \beta SI - \mu S) + \frac{(a_2 - 1)\sigma^2}{2} S^2 I^2 \right). \quad (17)$$

For any $S \in (0, 1 - \varepsilon_1]$, we have $b(S + 1)(1 - S) \geq \mu\varepsilon_1$, we have

$$(S + 1)(\mu - \beta SI - \mu S) + \frac{(a_2 - 1)\sigma^2}{2} S^2 I^2 \geq \frac{\mu\varepsilon_1}{2},$$

Hence, we have $\mathcal{L}V_3 \leq \frac{-\mu\varepsilon_1}{2}$. It then follows from Dynkin's formula [29] that

$$\mathbb{E}[V_3(S(\tau_{\varepsilon_1} \vee t), I(\tau_{\varepsilon_1} \vee t), R(\tau_{\varepsilon_1} \vee t))] \leq V_3(S_0, I_0, R_0) - \frac{\mu\varepsilon_1}{2} \mathbb{E}(\tau_{\varepsilon_1} \vee t). \quad (18)$$

Let $t \rightarrow \infty$, then by applying Fatou's lemma, we get

$$\mathbb{E}[V_3(S(\tau_{\varepsilon_1}), I(\tau_{\varepsilon_1}), R(\tau_{\varepsilon_1}))] \leq V_3(S_0, I_0, R_0) - \frac{\mu\varepsilon_1}{2} \mathbb{E}(\tau_{\varepsilon_1}). \quad (19)$$

Since V_3 is bounded on set Γ . Therefore, we obtain $E(\tau_{\varepsilon_1}) < \infty$. Finally, combining the strong Markov property, one obtains the assertion (8) that for any given $\varepsilon > 0$, and any $(S_0, I_0, R_0) \in \Gamma$. Hence, we get (7). This finishes our proof.

3.2 Persistence of the infectious disease

Theorem 3 *For any initial value $(S(0), I(0), R(0)) \in \Gamma$, if $\mathcal{R}_s > 1$, then the solution of the stochastic epidemic model (2) obeys*

1. $\liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t S(t) \geq \frac{\mu}{\mu + \beta}$
2. $\liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t I(t) \geq \frac{\mu\beta}{(\lambda + \mu + mu(t))(\beta + \mu)\mathcal{R}_0}(\mathcal{R}_s - 1)$
3. $\liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t R(t) \geq \frac{\lambda\mu\beta(\gamma + \mu)}{(\gamma + \mu)(\lambda + \mu + mu(t))(\beta + \mu)\mathcal{R}_0}(\mathcal{R}_s - 1)$

Proof We get the first assumption similarly as in [20].

Let consider the function V_2 defined on Γ such as

$$V_2(S, I, R) = \log(I) + \frac{\lambda\rho}{(\mu + \gamma)^2} \log(R) + \frac{\beta}{\mu}S + \left(\frac{\beta}{\mu} + 1\right)I, \quad (20)$$

Applying Itô's formula on V_3 , we obtain

$$dV_2 \geq \left[\left(\beta - \left(\mu + \lambda + mu(t) - \frac{\lambda\rho}{\mu + \gamma} + \frac{\sigma^2}{2} \right) \right) - (\lambda + \mu + mu(t)) \left(\frac{\beta}{\mu} + 1 \right) I \right] dt \\ [(1 + I)\sigma S] dB(t), \quad (21)$$

Since V_2 is bounded, by integrating (21), we obtain

$$\frac{1}{t} \int_0^t I(s)ds \geq \frac{\mu}{(\lambda + \mu + mu(t))(\beta + \mu)} \left[\frac{V_2(0) - V_2(t)}{t} + \frac{\beta}{\mathcal{R}_0}(\mathcal{R}_s - 1) + \frac{M(t)}{t} \right],$$

where $M(t) = \int_0^t (1 + I)\sigma S dB(s)$. Therefore, we get the second assertion by the Law of large numbers [21]. Finally using the second assertion and by integrating the third equation of (2) and dividing both sides by t we get the third assertion.

4 Near Optimality

The main idea of this work is to study a near-optimal for treatment strategy, which represents a more general approach instead of an optimal control [20] of the stochastic epidemic system (2). Hence, in this section, we establish sufficient and necessary conditions for near-optimal control.

4.1 Sufficient condition of near optimality

In this part, firstly we will handle some lemmas that will be applied to obtain sufficient conditions for stochastic near-optimal control. For convenience, let $x(t) = (S(t), I(t), R(t))^T = (x_1(t), x_2(t), x_3(t))^T$

Lemma 2 *For any $\alpha \geq 0$, if $(x(t), u(t))$ is an admissible pair, then*

$$\sum_{i=1}^3 \mathbb{E} \left\{ \sup_{0 \leq t \leq T} |x_i(t)|^\alpha \right\} \leq C, \quad (22)$$

with C is a constant that depends only on α .

Proof According to Theorem (1), then the system's processes are bounded, which means the inequality (22) holds.

Let $x(t) = (x_1(t), x_2(t), x_3(t))^\top$ is the corresponding state trajectory. Now we define the first-order adjoint solution $(p(t), q(t)) \in L^2_{\mathcal{F}}(0, T; \mathbb{R}^3) \times L^2_{\mathcal{F}}(0, T; \mathbb{R}^3)$ satisfying the following backward adjoint equation corresponding to (2) :

$$\begin{aligned} dp_1(t) &= -\Psi_1(x(t), u(t), p(t), q(t))dt + q_1(t)dB(t), \\ dp_2(t) &= -\Psi_2(x(t), u(t), p(t), q(t))dt + q_2(t)dB(t), \\ dp_3(t) &= -\Psi_3(x(t), u(t), p(t), q(t))dt, \\ p_i(T) &= g_{x_i}(x_i(T)), \quad i = 1, 2, 3. \end{aligned} \quad (23)$$

We will use the short notations for convenience:

$$(\Psi(x(t), u(t), p(t), q(t))) = \Psi_i(x, u, p, q), \quad i = 1, 2, 3),$$

where

$$\begin{cases} \Psi_1(x, u, p, q) = -\mu p_1 + \beta I(p_2 - p_1) + \sigma_1 I(q_2 - q_1) + l_S(x, u), \\ \Psi_2(x, u, p, q) = \beta S(p_2 - p_1) + (-\lambda - \mu u(t))(p_2 - p_3) - \mu p_2 + \sigma_1 S(q_2 - q_1) + l_I(x, u), \\ \Psi_3(x, u, p, q) = -\mu p_3 + \rho(p_2 - p_3) + l_R(x, u). \end{cases}$$

We introduce the Hamiltonian function $H : [0, T] \times \mathbb{R}_+^3 \times \mathcal{U}_{ad} \times \mathbb{R}_+^3 \times \mathbb{R}_+^2 \times \Omega \rightarrow \mathbb{R}$, by

$$H(t, x, u, p, q) = l(x, u) + b^\top(x, u)p + \sigma^\top(x)q. \quad (24)$$

Such that

$$b(x, u) = \begin{bmatrix} b_1(x) \\ b_2(x, u) \\ b_3(x, u) \end{bmatrix} \quad \text{and} \quad \sigma(x) = \begin{bmatrix} \sigma_1(x) \\ \sigma_2(x) \end{bmatrix}.$$

Lemma 3 For any $u(t) \in \mathcal{U}_{ad}$, the solution of adjoint equation verify the following property

$$\sum_{i=1}^3 \mathbb{E} \left\{ \sup_{0 \leq t \leq T} |p_i(t)|^2 \right\} + \sum_{i=1}^2 \mathbb{E} \left\{ \int_0^T |q_i(t)|^2 dt \right\} \leq C, \quad (25)$$

where C is a constant.

Proof Integrating both sides of the adjoint equation (23) from t to T , we get

$$p_i(t) + \int_t^T q_i(s)dB(s) = p_i(T) + \int_t^T \Psi_i(x(s), u(s), p(s), q(s))ds. \quad (26)$$

For all $t \geq 0$, by using (26), we can get

$$\begin{aligned} & \mathbb{E}|p_1(t)|^2 + \mathbb{E} \left(\int_t^T |q_1(s)|^2 ds \right) \\ & \leq C\mathbb{E}|p_1(T)|^2 + C(T-t)\mathbb{E} \left\{ \int_t^T |\Psi_1(x, u, p, q)|^2 ds \right\} \\ & \leq C\mathbb{E}|p_1(T)|^2 + C(T-t) \sum_{i=1}^2 \mathbb{E} \left(\int_t^T |p_i(s)|^2 ds \right) \\ & \quad + C(T-t) \sum_{i=1}^2 \mathbb{E} \left(\int_t^T |q_i(s)|^2 ds \right). \end{aligned} \quad (27)$$

Similarly, for the second equation, we have

$$\begin{aligned}
& \mathbb{E}|p_2(t)|^2 + \mathbb{E} \left(\int_t^T |q_2(s)|^2 ds \right) \\
& \leq C\mathbb{E}|p_2(T)|^2 + C(T-t)\mathbb{E} \left\{ \int_t^T |\Psi_2(x, u, p, q)|^2 ds \right\} \\
& \leq C\mathbb{E}|p_2(T)|^2 + C(T-t) \sum_{i=1}^3 \mathbb{E} \left(\int_t^T |p_i(s)|^2 ds \right) \\
& + C(T-t) \sum_{i=1}^2 \mathbb{E} \left(\int_t^T |q_i(s)|^2 ds \right). \tag{28}
\end{aligned}$$

Following the same method for the third equation, we can get

$$\begin{aligned}
\mathbb{E}|p_3(t)|^2 & \leq C\mathbb{E}|p_3(T)|^2 + C(T-t)\mathbb{E} \left\{ \int_t^T |\Psi_3(x, u, p, q)|^2 ds \right\} \\
& \leq C\mathbb{E}|p_3(T)|^2 + C(T-t) \sum_{i=2}^3 \mathbb{E} \left(\int_t^T |p_i(s)|^2 ds \right). \tag{29}
\end{aligned}$$

We add (27), (28) and (29) we can obtain that

$$\begin{aligned}
& \sum_{i=1}^3 \mathbb{E}|p_i(t)|^2 + \sum_{i=1}^2 \mathbb{E} \left(\int_t^T |q_i(s)|^2 ds \right) \\
& \leq C \sum_{i=1}^3 \mathbb{E}|p_i(T)|^2 + C(T-t) \sum_{i=1}^3 \mathbb{E} \left(\int_t^T |p_i(s)|^2 ds \right) \\
& + C(T-t) \sum_{i=1}^2 \mathbb{E} \left(\int_t^T |q_i(s)|^2 ds \right). \tag{30}
\end{aligned}$$

For $t \in [T - \varepsilon, T]$, with $\varepsilon = \frac{1}{2C}$, we will obtain

$$\begin{aligned}
& \sum_{i=1}^3 \mathbb{E}|p_i(t)|^2 + \frac{1}{2} \left\{ \sum_{i=1}^2 \mathbb{E} \left(\int_t^T |q_i(s)|^2 ds \right) \right\} \\
& \leq C \sum_{i=1}^3 \mathbb{E}|p_i(T)|^2 + C(T-t) \sum_{i=1}^3 \mathbb{E} \left(\int_t^T |p_i(s)|^2 ds \right)
\end{aligned}$$

By using Gronwall's inequality, we have

$$\sum_{i=1}^3 \sup_{0 \leq t \leq T} \mathbb{E} |p_i(t)|^2 \leq C, \quad \sum_{i=1}^2 \mathbb{E} \left(\int_t^T |q_i(s)|^2 ds \right) \leq C. \tag{31}$$

Using the same method we get the estimate (31) for all $t \in [T - 2\varepsilon, T]$.

Repeating the iteration for a finite integer n and $\forall t \in [T - n\varepsilon, T]$ until we obtain

that the statement holds for all $t \in [0, T]$.

The equation (26) can be rewritten as follows

$$p_i(t) = p_i(T) + \int_t^T \Psi_i(x(s), u(s), p(s), q(s)) ds - \int_0^T q_i(s) dB(s) + \int_0^t q_i(s) dB(s). \quad (32)$$

Using the elementary inequality

$$\left| \sum_{i=1}^4 a_i \right|^n \leq 4^n \left(\sum_{i=1}^4 |a_i|^n \right), \quad \forall n > 0. \quad (33)$$

We have

$$\begin{aligned} |p_1(t)|^2 &\leq C \left[|p_1(T)|^2 + \int_0^T \left(\sum_{i=1}^2 \{|p_i(s)|^2 + |q_i(s)|^2\} \right) ds \right. \\ &\quad \left. + \left(\int_0^T q_i(s) dB(s) \right)^2 + \left(\int_0^t q_i(s) dB(s) \right)^2 \right]. \end{aligned}$$

Following the same steps for the second and third equation, we can get

$$\begin{aligned} \sum_{i=1}^3 |p_i(t)|^2 &\leq C \left[\sum_{i=1}^3 |p_i(T)|^2 + \int_0^T \left(\sum_{i=1}^3 |p_i(s)|^2 + \sum_{i=1}^2 |q_i(s)|^2 \right) ds \right. \\ &\quad \left. + \sum_{i=1}^2 \left(\int_0^T q_i(s) dB(s) \right)^2 + \sum_{i=1}^2 \left(\int_0^t q_i(s) dB(s) \right)^2 \right]. \end{aligned}$$

Applying Burkholder–Davis–Gundy inequality, we obtain

$$\sum_{i=1}^3 \sup_{0 \leq t \leq T} \mathbb{E} |p_i(t)|^2 \leq C \left[\sum_{i=1}^3 \mathbb{E} |p_i(T)|^2 + \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T |p_i(s)|^2 ds \right\} + \sum_{i=1}^2 \mathbb{E} \left\{ \int_0^T |q_i(s)|^2 ds \right\} \right].$$

By using Gronwall's inequality, we can directly obtain our desired result (25). This completes the proof.

To introduce the sufficient conditions of near optimality for the stochastic model (2).

For any $\epsilon > 0$, we define a complete metric d' on $\mathcal{U}_{ad}[0, T]$ as follows:

$$d'(u(t), u'(t)) = \mathbb{E} \left\{ \int_0^T r^\epsilon(t) |u(t) - u'(t)| dt \right\}, \quad (34)$$

with

$$r^\epsilon(t) = 1 + \sum_{i=1}^3 |p_i^\epsilon(t)| + \sum_{i=1}^2 |q_i^\epsilon(t)|.$$

Theorem 4 *We suppose that (H_1) – (H_2) hold. Let (x^ϵ, u^ϵ) be admissible pair and $(p^\epsilon(t), q^\epsilon(t))$ be the solutions of adjoint equation (23), we assume that $H(t, x, u, p, q)$ and $g(x)$ are both convex, then for any $\epsilon > 0$, if*

$$\mathbb{E} \int_0^T H(t, x^\epsilon, u^\epsilon, p^\epsilon, q^\epsilon) dt \geq \sup_{u \in \mathcal{U}_{ad}[0, T]} \mathbb{E} \int_0^T H(t, x^\epsilon, u, p^\epsilon, q^\epsilon) dt - \epsilon. \quad (35)$$

Then

$$J(0, x_0, u^\epsilon) \leq \inf_{u \in \mathcal{U}_{ad}[0, T]} J(0, x_0, u) + C\epsilon^{\frac{1}{2}}. \quad (36)$$

Proof From the Hamiltonian function and the cost function (24), we obtain

$$J(0, x_0, u^\varepsilon) - J(0, x_0, u) = W_1 + W_2 - W_3. \quad (37)$$

where

$$\begin{aligned} W_1 &= E \int_0^T \{H(t, x^\varepsilon, u^\varepsilon, p^\varepsilon, q^\varepsilon) - H(t, x, u, p^\varepsilon, q^\varepsilon)\} dt, \\ W_2 &= E \{g(x^\varepsilon(T)) - g(x(T))\}, \\ W_3 &= E \int_0^T \{[b^\top(x^\varepsilon, u^\varepsilon) - b^\top(x, u)]p^\varepsilon + [\sigma^\top(x^\varepsilon) - \sigma^\top(x)]q^\varepsilon\} dt. \end{aligned}$$

Using the convexity of H and g , we can obtain

$$W_1 \leq \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T [H_{x_i}(x^\varepsilon, u^\varepsilon, p^\varepsilon, q^\varepsilon)(x_i^\varepsilon - x_i) + H_u(x^\varepsilon, u^\varepsilon, p^\varepsilon, q^\varepsilon)(u^\varepsilon - u)] dt \right\} \quad (38)$$

Similarly,

$$W_2 = \mathbb{E} [g(x^\varepsilon(T)) - g(x(T))] \leq \sum_{i=1}^3 \mathbb{E} \{g_{x_i}(x_i^\varepsilon(T)) [x_i^\varepsilon(T) - x_i(T)]\}. \quad (39)$$

Now, we use the Itô's formula to $\sum_{i=1}^3 p_i(t)^\varepsilon (x_i^\varepsilon(t) - x_i(t))$. We get

$$\begin{aligned} \sum_{i=1}^3 \mathbb{E} (g_{x_i}(x_i^\varepsilon(T)) [x_i^\varepsilon(T) - x_i(T)]) &= - \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T (x_i^\varepsilon(t) - x_i(t)) H_{x_i}(x^\varepsilon(t), u^\varepsilon(t), p^\varepsilon(t), q^\varepsilon(t)) dt \right\} \\ &+ \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T p_i^\varepsilon(t) [b_i(x^\varepsilon(t), u^\varepsilon(t)) - b_i(x(t), u(t))] dt \right\} + \sum_{i=1}^2 \mathbb{E} \left\{ \int_0^T q_i^\varepsilon(t) (\sigma_i(x^\varepsilon(t)) - \sigma_i(x(t))) dt \right\}. \quad (40) \end{aligned}$$

Then

$$\begin{aligned} W_3 &= \sum_{i=1}^3 \mathbb{E} (g_{x_i}(x_i^\varepsilon(T)) [x_i^\varepsilon(T) - x_i(T)]) \\ &+ \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T (x_i^\varepsilon(t) - x_i(t)) H_{x_i}(x^\varepsilon(t), u^\varepsilon(t), p^\varepsilon(t), q^\varepsilon(t)) dt \right\}. \quad (41) \end{aligned}$$

We combine (38) and (41), after we replace them in (37), we get

$$J(0, x_0, u^\varepsilon) - J(0, x_0, u) \leq \mathbb{E} \left\{ \int_0^T H_u(x^\varepsilon(t), u^\varepsilon(t), p^\varepsilon(t), q^\varepsilon(t)) (u^\varepsilon(t) - u(t)) dt \right\}. \quad (42)$$

To estimate the term $H_u(x^\varepsilon(t), u^\varepsilon(t), p^\varepsilon(t), q^\varepsilon(t)) (u^\varepsilon(t) - u(t))$, we define a functional

$$A(\cdot) : \mathcal{U}_{ad}[0, T] \rightarrow \mathbb{R}. \quad (43)$$

Such that

$$A(u(t)) = \mathbb{E} \left(\int_0^T H(x^\varepsilon(t), u(t), p^\varepsilon(t), q^\varepsilon(t)) dt \right). \quad (44)$$

By Hypothesis (H_2) , we prove that $A(u(t))$ is continuous on $\mathcal{U}_{ad}[0, T]$ with respect to the metric d' . Then by Ekeland's principle in lemma (1), there exists a $\bar{u}^\varepsilon(t) \in \mathcal{U}_{ad}[0, T]$ such that

$$d'(u^\varepsilon, \bar{u}^\varepsilon(t)) \leq \varepsilon^{\frac{1}{2}}, \quad \text{and} \quad A(\bar{u}^\varepsilon(t)) \leq A(u(t)) + \varepsilon^{\frac{1}{2}} d'(u(t), \bar{u}^\varepsilon(t)),$$

For all $u(t) \in \mathcal{U}_{ad}$. Thus,

$$H(x^\varepsilon(t), \bar{u}^\varepsilon(t), p^\varepsilon(t), q^\varepsilon(t)) = \min_{u \in \mathcal{U}_{ad}} \left(H(x^\varepsilon(t), u(t), p^\varepsilon(t), q^\varepsilon(t)) + \varepsilon^{\frac{1}{2}} r^\varepsilon(t) |u(t) - \bar{u}^\varepsilon(t)| \right).$$

According to [26], we can get

$$0 \in \partial_u H(x^\varepsilon(t), \bar{u}^\varepsilon(t), p^\varepsilon(t), q^\varepsilon(t)) \subset \partial_u H(x^\varepsilon(t), \bar{u}^\varepsilon(t), p^\varepsilon(t), q^\varepsilon(t)) + [-\varepsilon^{\frac{1}{2}} r^\varepsilon(t), \varepsilon^{\frac{1}{2}} r^\varepsilon(t)].$$

From (H_2) the Hamiltonian function H is differentiable in $u(t)$, then there exists $z^\varepsilon(t) \in [-\varepsilon^{\frac{1}{2}} r^\varepsilon(t), \varepsilon^{\frac{1}{2}} r^\varepsilon(t)]$, such that

$$H_u(x^\varepsilon(t), \bar{u}^\varepsilon(t), p^\varepsilon(t), q^\varepsilon(t)) + z^\varepsilon(t) = 0.$$

According to the assumption (H_2) , we get

$$\begin{aligned} |H_u(x^\varepsilon, u^\varepsilon(t), p^\varepsilon, q^\varepsilon)| &\leq |H_u(x^\varepsilon, u^\varepsilon(t), p^\varepsilon, q^\varepsilon) - H_u(x^\varepsilon, \bar{u}^\varepsilon(t), p^\varepsilon, q^\varepsilon)| \\ &\quad + |H_u(x^\varepsilon, \bar{u}^\varepsilon(t), p^\varepsilon, q^\varepsilon)| \\ &\leq Cr^\varepsilon(t) |u^\varepsilon(t) - \bar{u}^\varepsilon(t)| + z^\varepsilon(t) \\ &\leq Cr^\varepsilon(t) |u^\varepsilon(t) - \bar{u}^\varepsilon(t)| + 2\varepsilon^{\frac{1}{2}} r^\varepsilon(t). \end{aligned} \quad (45)$$

with $(H(x^\varepsilon, \bar{u}^\varepsilon(t), p^\varepsilon, q^\varepsilon) = H(x^\varepsilon(t), \bar{u}^\varepsilon(t), p^\varepsilon(t), q^\varepsilon(t))$

According to Lemma (3) and the definition of d' in (34), we can achieve the desired conclusion from (42), (45) and using Hölder's inequality we obtain (36), this completes the proof.

4.2 Necessary condition of near optimality

In this Section, we will provide the necessary condition for near-optimal controls of model (2). To get our results, we define a metric on $\mathcal{U}_{ad}[0, T]$ as follows

$$d(u(t), u'(t)) = \mathbb{E} [\Delta \{t \in [0, T] : u(t) \neq u'(t)\}]. \quad (46)$$

with Δ expresses Lebesgue measure. Following Lemma 6.4 in [22] to get that $\mathcal{U}_{ad}$ is a complete space under the metric d .

Lemma 4 *For any $n_1 \geq 0$ and $0 < n_2 < 1$ verifying $n_1 n_2 < 1$, there exists a constant $C = C(n_1, n_2)$ and for any $u(t), u'(t) \in \mathcal{U}_{ad}$ with the corresponding paths $x(t), x'(t)$, we have*

$$\sum_{i=1}^3 \mathbb{E} \left\{ \sup_{0 \leq t \leq T} |x_i(t) - x'_i(t)|^{2n_1} \right\} \leq C d(u(t), u'(t))^{n_1 n_2}. \quad (47)$$

Proof Case 1: $n_1 \geq 1$

First, we will estimate $|x_1(t) - x'_1(t)|^{2n_1}$. Let $\kappa > 1$ and $\eta = \frac{1}{n_1 n_2} > 1$ such that $\frac{1}{\kappa} + \frac{1}{\eta} = 1$, with for any $\theta > 0$. Then by using Cauchy-Schwartz's inequality and Burkholder–Davis–Gundy inequality, we can get

$$\begin{aligned} \mathbb{E} \left\{ \sup_{0 \leq t \leq \theta} |x_1(t) - x'_1(t)|^{2n_1} \right\} &\leq C \mathbb{E} \left(\int_0^\theta [\mu^{2n_1} |x_1(t) - x'_1(t)|^{2n_1} \right. \\ &\quad \left. + (\beta^{2n_1} + \sigma_1^{2n_1}) |x_1(t)x_2(t) - x'_1(t)x'_2(t)|^{2n_1}] dt \right), \\ &\leq C \mathbb{E} \left\{ \int_0^\theta \sum_{i=1}^2 |x_i(t) - x'_i(t)|^{2n_1} dt \right\}. \end{aligned} \quad (48)$$

Next, by the same argument to estimate $|x_2(t) - x'_2(t)|^{2n_1}$ we get

$$\begin{aligned} \mathbb{E} \left\{ \sup_{0 \leq t \leq \theta} |x_2(t) - x'_2(t)|^{2n_1} \right\} &\leq C \mathbb{E} \left(\int_0^\theta [(\beta^{2n_1} + \sigma_1^{2n_1}) |x_1(t)x_2(t) - x'_1(t)x'_2(t)|^{2n_1} \right. \\ &\quad \left. + \rho^{2n_1} |x_3(t) - x'_3(t)|^{2n_1} + (\mu^{2n_1} + \lambda_1^{2n_1}) |x_2(t) - x'_2(t)|^{2n_1} \right. \\ &\quad \left. + m^{2n_1} |u(t)x_2(t) - u'(t)x'_2(t)|^{2n_1}] dt \right) \\ &\leq C \mathbb{E} \left\{ \int_0^\theta \sum_{i=1}^3 |x_i(t) - x'_i(t)|^{2n_1} dt \right\} \\ &\quad + C \left\{ \mathbb{E} \left[\int_0^\theta 1^\kappa dt \right] \right\}^{\frac{1}{\kappa}} \left\{ \mathbb{E} \int_0^\theta \chi_{\{u(t) \neq u'(t)\}} dt \right\}^{\frac{1}{\eta}}, \\ &\leq C \mathbb{E} \left\{ \int_0^\theta \sum_{i=1}^3 |x_i(t) - x'_i(t)|^{2n_1} dt \right\} + C \left\{ \mathbb{E} \int_0^\theta \chi_{\{u(t) \neq u'(t)\}} dt \right\}^{n_1 n_2}, \\ &\leq C \left\{ \mathbb{E} \left[\int_0^\theta \sum_{i=1}^3 |x_i(t) - x'_i(t)|^{2n_1} dt \right] + d(u(t), u'(t))^{n_1 n_2} \right\}. \end{aligned} \quad (49)$$

Similarly to estimate $|x_3(t) - x'_3(t)|^{2n_1}$, we have

$$\mathbb{E} \left\{ \sup_{0 \leq t \leq \theta} |x_3(t) - x'_3(t)|^{2n_1} \right\} \leq C \left\{ \mathbb{E} \left[\int_0^\theta \sum_{i=2}^3 |x_i(t) - x'_i(t)|^{2n_1} dt \right] + d(u(t), u'(t))^{n_1 n_2} \right\}. \quad (50)$$

Adding up (48), (49) and (50), we have

$$\sum_{i=1}^3 \mathbb{E} \left\{ \sup_{0 \leq t \leq \theta} |x_i(t) - x'_i(t)|^{2n_1} \right\} \leq C \left\{ \mathbb{E} \left[\int_0^\theta \sum_{i=1}^3 \sup_{0 \leq t \leq \theta} |x_i(t) - x'_i(t)|^{2n_1} dt \right] + d(u(t), u'(t))^{n_1 n_2} \right\}.$$

From the above inequalities and Gronwall inequality, we obtain directly the conclusion.

Case 2: $0 \leq n_1 \leq 1$

By using Cauchy-Schwartz inequality, we can obtain

$$\begin{aligned} \sum_{i=1}^3 \mathbb{E} \left\{ \sup_{0 \leq t \leq \theta} |x_i(t) - x'_i(t)|^{2n_1} \right\} &\leq \sum_{i=1}^3 \left\{ \mathbb{E} \sup_{0 \leq t \leq \theta} |x_i(t) - x'_i(t)|^2 \right\}^{n_1} \\ &\leq \{Cd(u(t), u'(t))^{n_2}\}^{n_1} \\ &= Cd(u(t), u'(t))^{n_1 n_2} \end{aligned}$$

This completes the proof.

Lemma 5 *Let conditions (H_1) and (H_2) hold. For any $1 < n_1 < 2$ and $0 < n_2 < 1$ satisfying $(1 + n_2)n_1 < 2$, there exists a constant $C = C(n_1, n_2)$ such that for any $u(t), u'(t) \in \mathcal{U}_{ad}$ along with the corresponding trajectory $x(t), x'(t)$ and the solution $(p(t), q(t)), (p'(t), q'(t))$ of corresponding adjoint equation, we have*

$$\sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T |p_i(t) - p'_i(t)|^{n_1} dt \right\} + \sum_{i=1}^2 \mathbb{E} \left\{ \int_0^T |q_i(t) - q'_i(t)|^{n_2} dt \right\} \leq C d(u(t), u'(t))^{\frac{n_1 n_2}{2}}. \quad (51)$$

Proof Let $\hat{p}_i(t) = p_i(t) - p'_i(t)$, and $\hat{q}_i(t) = q_i(t) - q'_i(t)$ for $(i = 1, 2, 3)$. According to the adjoint equation system (23), we obtain

$$\begin{aligned} d\hat{p}_1(t) &= -[-\mu\hat{p}_1(t) + \beta I(t)(\hat{p}_2 - \hat{p}_1)(t) + \sigma_1 I(t)(\hat{q}_2 - \hat{q}_1)(t) + \psi_1(t)]dt + \hat{q}_1(t)dB(t) \\ d\hat{p}_2(t) &= -[\beta S(t)(\hat{p}_2 - \hat{p}_1)(t) + (-\lambda - \mu u(t))(\hat{p}_2 - \hat{p}_3)(t) - \mu\hat{p}_2(t) + \sigma_1 S(t)(\hat{q}_2 - \hat{q}_1)(t) + \psi_2(t)]dt \\ &\quad + \hat{q}_2(t)dB(t), \\ d\hat{p}_3(t) &= -[-\mu\hat{p}_3(t) + (\rho)(\hat{p}_2 - \hat{p}_3)(t) + \psi_3(t)]dt \\ \hat{p}_i(T) &= g_{x_i}(x_i(T)) - g_{x_i}(x'_i(T)), \quad (i = 1, 2, 3.) \end{aligned} \quad (52)$$

with

$$\begin{aligned} \psi_1(t) &= \beta(p'_2 - p'_1)(t)(I - I')(t) + \sigma_1(q'_2 - q'_1)(t)(I - I')(t) + l_S(x, u) - l_S(x', u'), \\ \psi_2(t) &= \beta(p'_2 - p'_1)(t)(S - S')(t) - m(p'_2 - p'_3)(t)(u(t) - u'(t)) + \sigma_1(q'_2 - q'_1)(t)(S - S')(t) \\ &\quad + l_I(x, u) - l_I(x', u'), \\ \psi_3(t) &= l_R(x, u) - l_R(x', u'). \end{aligned} \quad (53)$$

We suppose that $\varphi(t) = (\varphi_1(t), \varphi_2(t), \varphi_3(t))^\top$ is the solution of the following linear forward stochastic differential equation .

$$\begin{aligned} d\varphi_1(t) &= \left\{ [-\mu - \beta I(t)]\varphi_1(t) - \beta S(t)\varphi_2(t) + |\hat{p}_1(t)|^{n_1-1} \text{sgn}(\hat{p}_1(t)) \right\} dt \\ &\quad + \left\{ [-\sigma_1 I(t)\varphi_1(t) - \sigma_1 S(t)\varphi_2(t) + |\hat{q}_1(t)|^{n_1-1} \text{sgn}(\hat{q}_1(t))] \right\} dB(t), \\ d\varphi_2(t) &= \left\{ \beta I(t)\varphi_1(t) + [\beta S(t) - (\mu + \lambda) - \mu u(t)]\varphi_2(t) + \rho\varphi_3(t) \right. \\ &\quad \left. + |\hat{p}_2(t)|^{n_1-1} \text{sgn}(\hat{p}_2(t)) \right\} dt \\ &\quad + \left\{ [\sigma_1 I(t)\varphi_1(t) + \sigma_1 S(t)\varphi_2(t) + |\hat{q}_2(t)|^{n_1-1} \text{sgn}(\hat{q}_2(t))] \right\} dB(t), \\ d\varphi_3(t) &= \left\{ [\lambda + \mu u(t)]\varphi_3(t) - [\mu + \rho]\varphi_3(t) + |\hat{p}_3(t)|^{n_1-1} \text{sgn}(\hat{p}_3(t)) \right\} dt, \\ \varphi(0) &= 0, \quad i = 1, 2, 3, \end{aligned} \quad (54)$$

where $sgn(\cdot)$ is a symbolic function. We obtain the existence and uniqueness of solutions of the system (54) by using the hypothesis (H_1) and lemma(4). For $n_3 > 2$ such that $\frac{1}{n_1} + \frac{1}{n_3} = 1$. From Cauchy-Schwarz inequality, we get

$$\sum_{i=1}^3 \mathbb{E} \left\{ \sup_{0 \leq t \leq T} |\varphi_i(t)|^{n_3} \right\} \leq \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T |\hat{p}_i(t)|^{n_1} dt \right\} + \sum_{i=1}^2 \mathbb{E} \left\{ \int_0^T |\hat{q}_i(t)|^{n_1} dt \right\}.$$

Applying Itô's formula on $\sum_{i=1}^3 \hat{p}_i(t)\varphi_i(t)$ and taking expectations, we have

$$\begin{aligned} & \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T |\hat{p}_i(t)|^{n_1} dt \right\} + \sum_{i=1}^2 \mathbb{E} \left\{ \int_0^T |\hat{q}_i(t)|^{n_1} dt \right\} \\ &= \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T \psi_i(t) \varphi_i(t) dt \right\} + \sum_{i=1}^3 \varphi_i(T) [g_{x_i}(x(T)) - g_{x_i}(x'(T))] \\ &\leq C \left[\sum_{i=1}^3 \mathbb{E} \left(\int_0^T |\psi_i(t)|^{n_1} dt \right)^{\frac{1}{n_1}} \mathbb{E} \left(\int_0^T |\varphi_i(t)|^{n_3} dt \right)^{\frac{1}{n_3}} \right. \\ &\quad \left. + \sum_{i=1}^3 (\mathbb{E} |g_{x_i}(x(T)) - g_{x_i}(x'(T))|^{n_1})^{\frac{1}{n_1}} (\mathbb{E} |\varphi_i(T)|^{n_3})^{\frac{1}{n_3}} \right] \\ &\leq C \left[\sum_{i=1}^3 \mathbb{E} \int_0^T |\hat{p}_i(t)|^{n_1} dt + \sum_{i=1}^2 \mathbb{E} \int_0^T |\hat{q}_i(t)|^{n_1} dt \right]^{\frac{1}{n_3}} \\ &\quad \times \left[\left(\sum_{i=1}^3 \mathbb{E} \int_0^T |\psi_i(t)|^{n_1} dt \right)^{\frac{1}{n_1}} + \left(\sum_{i=1}^3 \mathbb{E} |g_{x_i}(x(T)) - g_{x_i}(x'(T))|^{n_1} \right)^{\frac{1}{n_1}} \right]. \end{aligned}$$

Using elementary inequality, we can get

$$\begin{aligned} & \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T |\hat{p}_i(t)|^{n_1} dt \right\} + \sum_{i=1}^2 \mathbb{E} \left\{ \int_0^T |\hat{q}_i(t)|^{n_1} dt \right\} \\ &\leq C \left[\sum_{i=1}^3 \mathbb{E} \int_0^T |\psi_i(t)|^{n_1} dt + \sum_{i=1}^3 \mathbb{E} |g_{x_i}(x(T)) - g_{x_i}(x'(T))|^{n_1} \right]. \end{aligned} \quad (55)$$

Now, to estimate the right-hand side of (55), we use (H_2) and lemma(4), we have

$$\sum_{i=1}^3 \mathbb{E} |g_{x_i}(x(T)) - g_{x_i}(x'(T))|^{n_1} \leq C^{n_1} \sum_{i=1}^3 \mathbb{E} |x_i(T) - x'_i(T)|^{n_1} \leq C \sum_{i=1}^3 d(u(t), u'(t))^{\frac{n_1 n_2}{2}}. \quad (56)$$

According to $\frac{n_1 n_2}{2} < 1 - \frac{n_1}{2} < 1$, and applying Cauchy-Schwartz inequality, we get

$$\begin{aligned} \mathbb{E} \int_0^T |\psi_1(t)|^{n_1} dt &\leq C \mathbb{E} \left(\int_0^T |x_2(t) - x'_2(t)|^{n_1} \times \left[\sum_{i=1}^2 |p'_i(t)|^{n_1} + \sum_{i=1}^2 |q'_i(t)|^{n_1} \right] \right. \\ &\quad \left. + \int_0^T |l_{x_1}(x(t), u(t)) - l_{x_1}(x'(t), u'(t))|^{n_1} dt \right) \\ &\leq C \left(\mathbb{E} \int_0^T |x_2(t) - x'_2(t)|^{\frac{2n_1}{2+n_1}} dt \right)^{1-\frac{n_1}{2}} \left[\left(\sum_{i=1}^2 \mathbb{E} \int_0^T |p'_i(t)|^2 dt \right)^{\frac{n_1}{2}} \right. \\ &\quad \left. + \left(\sum_{i=1}^2 \mathbb{E} \int_0^T |q'_i(t)|^2 dt \right)^{\frac{n_1}{2}} \right] + d(u(t), u'(t))^{\frac{n_1 n_2}{2}}. \end{aligned}$$

The first right term is bounded by $Cd(u(t), u'(t))^{\frac{n_1 n_2}{2}}$. Noting that $\frac{2n_1}{1-n_1} < 1$. By (4) and (25), we get

$$\mathbb{E} \left[\int_0^T |\psi_1(t)|^{n_1} dt \right] \leq Cd(u(t), u'(t))^{\frac{n_1 n_2}{2}}. \quad (57)$$

For the next estimation, by using the system (53), we will get

$$\mathbb{E} \left[\int_0^T |\psi_2(t)|^{n_1} dt \right] \leq Cd(u(t), u'(t))^{\frac{n_1 n_2}{2}}. \quad (58)$$

Similarly, we have

$$\mathbb{E} \left[\int_0^T |\psi_3(t)|^{n_1} dt \right] \leq Cd(u(t), u'(t))^{\frac{n_1 n_2}{2}}. \quad (59)$$

Combining (57), (58) and (59), so we can get

$$\sum_{i=1}^3 \mathbb{E} \left[\int_0^T |\psi_i|^{n_1} dt \right] \leq Cd(u(t), u'(t))^{\frac{n_1 n_2}{2}}. \quad (60)$$

Next, adding (56) and (60) into (55), we get

$$\sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T |\hat{p}_i(t)|^{n_1} dt \right\} + \sum_{i=1}^2 \mathbb{E} \left\{ \int_0^T |\hat{q}_i(t)|^{n_1} dt \right\} \leq Cd(u(t), u'(t))^{\frac{n_1 n_2}{2}}.$$

This completes the proof.

Theorem 5 Suppose that (H_1) and (H_2) hold and g convex. $(p^\varepsilon(t), q^\varepsilon(t))$ be the solution of the adjoint equation (23). For any $n_2 \in [0, 1)$, $\varepsilon > 0$ and any ε -optimal pair $(x^\varepsilon(t), u^\varepsilon(t))$, then there exist a constant C such that

$$\mathbb{E} \left\{ \int_0^T mI^\varepsilon(t)(p_3^\varepsilon(t) - p_2^\varepsilon(t))(u(t) - u^\varepsilon(t)) + l(x^\varepsilon(t), u(t)) - l(x^\varepsilon(t), u^\varepsilon(t)) dt \right\} \geq -C\varepsilon^{\frac{n_2}{3}}. \quad (61)$$

with C as a constant.

Proof From (H_1) the objective function $J(0, x_0, u(t)) : \mathcal{U}_{ad} \rightarrow \mathbb{R}$ is continuous under the metric d , by applying Ekelands principle and choosing $\lambda = \varepsilon^{\frac{2}{3}}$. then there exists an admissible pair $(\hat{x}^\varepsilon(t), \hat{u}^\varepsilon(t))$ such that

$$d(u^\varepsilon(t), \hat{u}^\varepsilon(t)) \leq \varepsilon^{\frac{2}{3}}, \quad (62)$$

and

$$\hat{J}(0, x_0, \hat{u}^\varepsilon(t)) \leq \hat{J}(0, x_0, u(t)) \quad \text{for all } u(t) \in \mathcal{U}_{ad}[0, T], \quad (63)$$

with

$$\hat{J}(0, x_0, u(t)) = J(0, x_0, u(t)) + \varepsilon^{\frac{1}{3}} d(u(t), \hat{u}^\varepsilon(t)). \quad (64)$$

This suggests that $(\hat{x}^\varepsilon(t), \hat{u}^\varepsilon(t))$ is an optimal pair for the system (2). To give the necessary condition we use a spike variation.

Let $\hat{t} \in [0, T]$, $\varrho > 0$ and $u(t) \in \mathcal{U}_{ad}[0, T]$, we define $u^\varrho(t) \in \mathcal{U}_{ad}[0, T]$ as follows

$$u^\varrho(t) = \begin{cases} u(t), & \text{if } t \in [\hat{t}, \hat{t} + \varrho], \\ \hat{u}^\varepsilon(t), & \text{if } t \in [0, T] \setminus [\hat{t}, \hat{t} + \varrho]. \end{cases}$$

Therefore, according to (62) and (63)

$$J(0, x_0, \hat{u}^\varepsilon(t)) \leq \hat{J}(0, x_0, u^\varrho(t)), \quad \text{and} \quad d(u^\varrho(t), \hat{u}^\varepsilon(t)) \leq \varrho. \quad (65)$$

On the basis of (65)-(64), lemma(4) and Taylor's expansion, we get

$$\begin{aligned} -\varrho \varepsilon^{\frac{1}{3}} &\leq J(0, x_0, u^\varrho(t)) - J(0, x_0, \hat{u}^\varepsilon(t)) \\ &= \mathbb{E} \int_0^T [l(x^\varrho(t), u^\varrho(t)) - l(\hat{x}^\varepsilon(t), \hat{u}^\varepsilon(t))] dt + \mathbb{E}[g(x^\varrho(T)) - g(\hat{x}^\varepsilon(T))] \\ &\leq \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T l_{x_i}(\hat{x}^\varepsilon(t), u^\varrho(t)) (x_i^\varrho(t) - \hat{x}^\varepsilon(t)) dt \right\} + \mathbb{E} \left\{ \int_{\hat{t}}^{\hat{t}+\varrho} [l(\hat{x}^\varepsilon(t), u(t)) - l(\hat{x}^\varepsilon(t), \hat{u}^\varepsilon(t))] dt \right\} \\ &\quad + \sum_{i=1}^3 [g_{x_i}(x^\varrho(T)) (x_i^\varrho(T) - \hat{x}_i^\varepsilon(T))] + o(\varrho). \end{aligned} \quad (66)$$

Using Itô's formula to $\sum_{i=1}^3 \hat{p}_i^\varepsilon(t) (x_i^\varrho(t) - \hat{x}_i^\varepsilon(t))$. From lemma (2) and (3), we have

$$\begin{aligned} &\sum_{i=1}^3 g_{x_i}(x^\varrho(T)) (x_i^\varrho(T) - \hat{x}_i^\varepsilon(T)) \\ &\leq \mathbb{E} \{ (-m[u^\varrho(t)x_2^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_2^\varepsilon(t)]\hat{p}_2^\varepsilon(t) + m[u^\varrho(t)x_3^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_3^\varepsilon(t)]\hat{p}_3^\varepsilon(t) \} \\ &\quad - \sum_{i=1}^3 \mathbb{E} \left\{ \int_0^T l_{x_i}(\hat{x}^\varepsilon(t), u^\varrho(t)) (x_i^\varrho(t) - \hat{x}_i^\varepsilon(t)) dt \right\}. \end{aligned} \quad (67)$$

Including (65) and replacing (67) into (66), then we can get

$$\begin{aligned} -\varrho \varepsilon^{\frac{1}{3}} &\leq J(0, x_0, u^\varrho(t)) - J(0, x_0, \hat{u}^\varepsilon(t)) \leq \mathbb{E} \left\{ \int_{\hat{t}}^{\hat{t}+\varrho} [l(\hat{x}^\varepsilon(t), u(t)) - l(\hat{x}^\varepsilon(t), \hat{u}^\varepsilon(t))] dt \right\} \\ &\quad + \mathbb{E} \left\{ \int_{\hat{t}}^{\hat{t}+\varrho} (-m[u^\varrho(t)x_2^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_2^\varepsilon(t)]\hat{p}_2^\varepsilon(t) + m[u^\varrho(t)x_3^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_3^\varepsilon(t)]\hat{p}_3^\varepsilon(t) dt \right\} \\ &\quad + o(\varrho). \end{aligned} \quad (68)$$

Dividing (68) by ϱ and tending $\varrho \rightarrow 0$, we obtain the following estimation

$$-\varepsilon^{\frac{1}{3}} \leq \mathbb{E}[l(\hat{x}^\varepsilon(t), u(t)) - l(\hat{x}^\varepsilon(t), \hat{u}^\varepsilon(t))] + \mathbb{E} \left\{ (-m[u^\varrho(t)x_2^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_2^\varepsilon(t)]\hat{p}_2^\varepsilon(t) + m[u^\varrho(t)x_3^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_3^\varepsilon(t)]\hat{p}_3^\varepsilon(t))dt \right\} \quad (69)$$

We can derive an estimate similar to the right hand of (68) with all $(\hat{x}^\varepsilon(t), \hat{u}^\varepsilon(t))$ replaced by $(x^\varepsilon(t), u^\varepsilon(t))$. Then, we estimate the following difference:

$$\begin{aligned} & \mathbb{E} \left\{ \int_0^T m[(u^\varrho(t)x_2^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_2^\varepsilon(t))\hat{p}_2^\varepsilon(t) - m[u^\varrho(t)x_2^\varrho(t) - u^\varepsilon(t)x_2^\varepsilon(t)]p_2^\varepsilon(t)]dt \right\} \\ & + \mathbb{E} \left\{ \int_0^T m[(u^\varrho(t)x_3^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_3^\varepsilon(t))\hat{p}_3^\varepsilon(t) - m[u^\varrho(t)x_3^\varrho(t) - u^\varepsilon(t)x_3^\varepsilon(t)]p_3^\varepsilon(t)]dt \right\} \\ & = \mathbb{E} \left\{ \int_0^T m(\hat{p}_2^\varepsilon(t) - p_2^\varepsilon(t))[u^\varrho(t)x_2^\varrho(t) - u^\varepsilon(t)x_2^\varepsilon(t)]dt \right\} + \mathbb{E} \left\{ \int_0^T m\hat{p}_2^\varepsilon(t)[u^\varepsilon(t)x_2^\varepsilon(t) - \hat{u}^\varepsilon(t)\hat{x}_2^\varepsilon(t)]dt \right\} \\ & + \mathbb{E} \left\{ \int_0^T m(\hat{p}_3^\varepsilon(t) - p_3^\varepsilon(t))[u^\varrho(t)x_3^\varrho(t) - u^\varepsilon(t)x_3^\varepsilon(t)]dt \right\} + \mathbb{E} \left\{ \int_0^T m\hat{p}_3^\varepsilon(t)[u^\varepsilon(t)x_3^\varepsilon(t) - \hat{u}^\varepsilon(t)\hat{x}_3^\varepsilon(t)]dt \right\} \\ & = L_1 + L_2 + L_3 + L_4. \end{aligned} \quad (70)$$

From lemma (2), lemma (5) and to $d(u^\varepsilon(t), \hat{u}^\varepsilon(t)) \leq \varepsilon^{\frac{2}{3}}$, we have for any $0 < n_2 < 1$, $1 < n_1 < 2$ where $(1 + n_1)n_2 < 2$, we can obtain

$$\begin{aligned} L_1 & \leq m \left(\mathbb{E} \int_0^T |\hat{p}_2^\varepsilon(t) - p_2^\varepsilon(t)|^{n_1} dt \right)^{\frac{1}{n_1}} \left(\mathbb{E} \int_0^T |u^\varrho(t)x_2^\varrho(t) - u^\varepsilon(t)x_2^\varepsilon(t)|^{\frac{n_1}{n_1-1}} dt \right)^{\frac{n_1-1}{n_1}} \\ & \leq C[d(u^\varepsilon(t), \hat{u}^\varepsilon(t))]^{\frac{n_1 n_2}{2}} \left(\mathbb{E} \int_0^T |u^\varrho(t)x_2^\varrho(t)|^{\frac{n_1}{n_1-1}} + |u^\varepsilon(t)x_2^\varepsilon(t)|^{\frac{n_1}{n_1-1}} dt \right)^{\frac{n_1-1}{n_1}} \\ & \leq C\varepsilon^{\frac{n_2}{3}}. \end{aligned} \quad (71)$$

Also

$$\begin{aligned} L_2 & \leq C \left(\mathbb{E} \int_0^T |\hat{p}_2^\varepsilon(t)|^2 dt \right)^{\frac{1}{2}} \left(\mathbb{E} \int_0^T |u^\varepsilon(t) - \hat{u}^\varepsilon(t)|^2 \chi_{\{u^\varepsilon(t) \neq \hat{u}^\varepsilon(t)\}}(t) dt \right)^{\frac{1}{2}} \\ & \leq C \left(\mathbb{E} \int_0^T |u^\varepsilon(t)|^4 + |\hat{u}^\varepsilon(t)|^4 dt \right)^{\frac{1}{4}} \left(\mathbb{E} \int_0^T \chi_{\{u^\varepsilon(t) \neq \hat{u}^\varepsilon(t)\}}(t) dt \right)^{\frac{1}{4}} \\ & \leq C d(u^\varepsilon(t), \hat{u}^\varepsilon(t))^{\frac{1}{4}} \\ & \leq C\varepsilon^{\frac{n_2}{3}}. \end{aligned} \quad (72)$$

In the same way we get the estimation to L_3 and L_4 and By a similar argument as (72), we have

$$\begin{aligned} & \mathbb{E} \left\{ \int_0^T m[(u^\varrho(t)x_2^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_2^\varepsilon(t))\hat{p}_2^\varepsilon(t) - m[u^\varrho(t)x_2^\varrho(t) - u^\varepsilon(t)x_2^\varepsilon(t)]p_2^\varepsilon(t)]dt \right. \\ & \left. + \int_0^T m[(u^\varrho(t)x_3^\varrho(t) - \hat{u}^\varepsilon(t)\hat{x}_3^\varepsilon(t))\hat{p}_3^\varepsilon(t) - m[u^\varrho(t)x_3^\varrho(t) - u^\varepsilon(t)x_3^\varepsilon(t)]p_3^\varepsilon(t)]dt \right\} \\ & \leq C\varepsilon^{\frac{n_2}{3}}. \end{aligned} \quad (73)$$

Similarly, we obtain

$$\begin{aligned} & \mathbb{E} \left\{ \int_0^T [l(\hat{x}^\varepsilon(t), u^\varepsilon(t)) - l(\hat{x}^\varepsilon(t), \hat{u}^\varepsilon(t))] - [l(x^\varepsilon(t), u^\varepsilon(t)) - l(x^\varepsilon(t), u^\varepsilon(t))] dt \right\} \\ & \leq C\varepsilon^{\frac{n_2}{3}}. \end{aligned} \quad (74)$$

Combining (69), (72), (73) and (74) and use the Hamiltonian function (24) we can get (61). This finishes the proof.

5 Characterization and numerical simulation of an optimal control

In this section, we formulate a characterization of a stochastic optimal control for a treatment strategy incorporated in the epidemic model (2) with relapse. Hence, in the following we discuss an optimal treatment $u^*(t)$ to achieve the minimization of the infected population according to an objective functional [27]. Moreover, the decrease of the infected size with a minimum pharmaceutical cost for which an initial state x_0 is determined as

$$E_{0,x_0} \left[\int_0^T \left(A_1 I(t) + A_2 \frac{u^2(t)}{2} \right) ds \right].$$

We consider that there is a positive constant $u^m \leq 1$ such that $u(t) \leq u^m$ a.s. Hence, The class of admissible control law is described as

$$\mathcal{A} = \{u(\cdot) \text{ is adapted} \mid 0 \leq u(t) \leq u^m \text{ a.s.}\}. \quad (75)$$

Therefore, to formulate a solution of this stochastic treatment problem, we define the performance criterion of the following form

$$\mathcal{J}(t, x, u) = E_{t,x} \left[\int_0^T (A_1 I(t) + A_2 \frac{u^2(t)}{2}) ds \right], \quad (76)$$

where the expectation is conditional on the state of the system being a fixed value at x at time t . The value function and the treatment control $u^* \in \mathcal{A}$ obtained by

$$\Phi(t, x) = \inf_{u(\cdot) \in \mathcal{A}} \mathcal{J}(t, x, u) = \mathcal{J}_s(t, x, u^*), \quad (77)$$

where we define a control law that minimizes the value $\mathcal{J} : \mathcal{A} \rightarrow \mathbb{R}_+$ given by (76). To resolve our stochastic optimal control problem for the stochastic system (2) given $\mathcal{A}$ in (75) and $\mathcal{J}$ in (76). We shall determine the value of the function Φ and a control function

$$u^* = \arg \inf_{u \in \mathcal{A}} \mathcal{J}_s(x; u(t)) \in \mathcal{A}. \quad (78)$$

For that matter, in the following theorem, we formulate the optimal control $u^*(t)$ for the pharmaceutical interventions.

Theorem 6 *A solution to the treatment receiving problem stated in Problem (75) is of the form*

$$u^*(t) = \max \left\{ \min \left\{ u^m, \frac{m(\Phi_I - \Phi_R)I(t)}{A_2} \right\}, 0 \right\}. \quad (79)$$

Proof The proof is similar to Theorem 7 in [20].

Next, we conduct a numerical analysis using Euler-Maruyama forward-backward approximation to simulate two scenarios with and without control for the stochastic systems (1) and (2) with relapse. Therefore, to perform those illustrations we adopt the following values for the parameters: $\mu = 0.07$, $\beta = 0.17$, $\sigma = 0.4$, $\lambda = 0.2$, $\rho = 0.7$ and $m = 1$, with the initial values $(S(0), I(0), R(0)) = (0.7, 0.2, 0.1)$. As result, the scenario in figure (1) simulates an infectious disease with relapse for 60 days, where its basic reproduction number is $\mathcal{R}_0 = 1.9278$. In the first days of the epidemic, we observe a wave of non-controlled systems, which affects nearly 70 percent of the population. Since the disease reverts back, even at this level of infected size, herd immunity cannot be built in the time of an epidemic. Therefore, a control treatment strategy is adopted to reduce the number of infected individuals. As a consequence, the more we reduce the infected population, the less we have as direct contact with susceptible individuals, which led to a decrease of nearly 50 percent in the infected size. However, to maintain this low level of infected individuals. It is suitable to exploit this policy with pharmaceutical interventions at its full limit when the number of infected is low in order to avoid some future peaks.

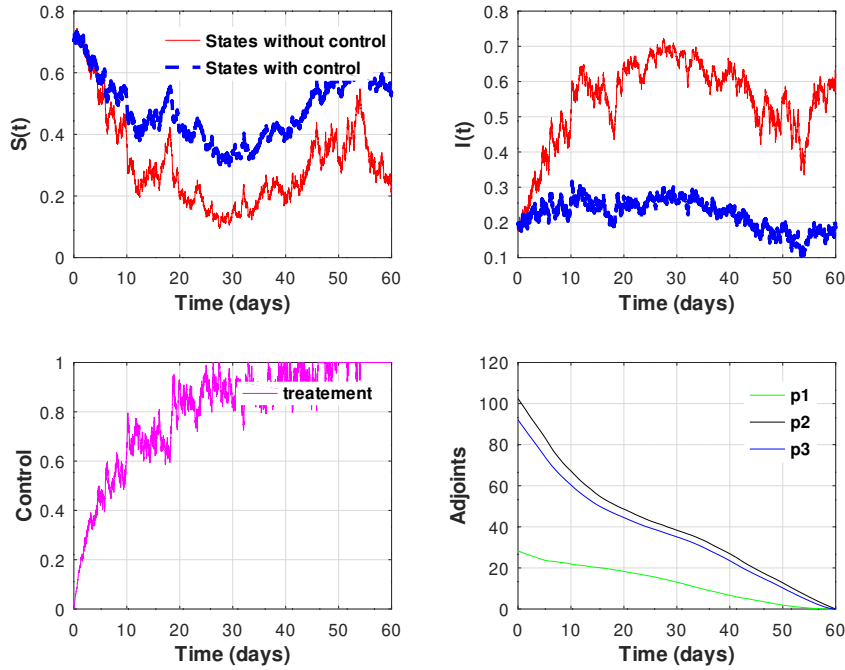


Fig. 1 Trajectories of stochastic epidemic model (2)

6 Sensitivity Analysis

In the literature, a sensitivity analysis is conducted to discuss the impact of the parameters of a dynamical system on the spread of infectious diseases. The main idea of this approach is to determine the ability of an epidemic model to resist changes without adjusting its initial formulation, where it plays a significant role to be aware, detecting, and acting on errors in the collected data to estimate a parameter value [28]. The most common sensitivity analysis approaches are the uncertainty in the estimation of infectious disease model parameters. Therefore, using a sensitivity index as a measure, we can reveal how much a variable is sensitive to a parameter in the model.

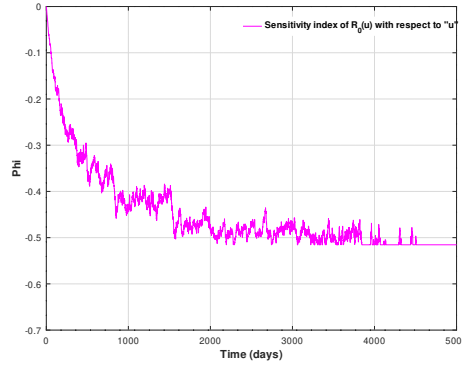


Fig. 2 normalized forward sensitivity index of the treatment policy to the basic reproduction number

In the figure (2), we illustrate a numerical approximation for a normalized forward sensitivity index of the optimal control for a treatment strategy, denoted by $\phi_{\mathcal{R}_0}^u$, to the stochastic epidemic model (2) with relapse. We observe that the index fluctuates in the negative sense, which means an increase in the treatment policy yields a proportional decrease in the basic reproduction rate. Therefore, full use of an optimal control treatment leads to high sensitivity. Since a slight shift in a highly sensitive parameter will result in significant quantitative changes, it should be carefully calculated.

7 Conclusion

In this paper, we try to answer the following question: Can a treatment strategy impacts reverted back infectious disease transmission in a population? Therefore, we establish a control variable to incorporate pharmaceutical interventions. Besides, to consider uncertainties according to environmental exposure, we include a white noise perturbation on the transmission rate. Hence, this work investigates stability analysis and near-optimal control problem of a stochastic SIRI [3] epidemic model for a treatment strategy. We provided a stochastic threshold for the

extinction and persistence in mean of an infectious disease with relapse and we established the sufficient and necessary conditions of near optimality. Next, we discuss a numerical characterization of a stochastic optimal control corresponding to a treatment strategy to reduce the size of the infected population graphically. After, we conduct a sensitivity analysis to measure the impact of the control variable on the basic reproduction number. However, in the absence of longitudinal data on diseases with relapse such as tuberculosis, our examples were not derived from real data but only to illustrate the various scenarios and to discuss the impact of such measures. Therefore, the estimation of parameters statistically for a real scenario is left for further investigation.

Data availability statement:

Data sharing is not applicable – no new data generated.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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